

Violence Against Civilians and Leader Favorability in Mali

Commune-level ACLED events and a three-wave, two-stage household survey

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1 Question

Is violence against civilians in a respondent's commune associated with how favorably they view the national leader? Violence is the number of ACLED events of violence against civilians in the commune in the period before each survey wave. Favorability comes from three survey waves.

2 Data and design

The analysis file has 5,036 respondents (165 dropped for missing data). Each wave is an independent two-stage sample within regions: communes are the primary sampling units and villages are sampled within communes. Because each wave is its own design, region, commune, and village codes are made wave-specific, and weights are rescaled to sum to each wave's sample size, $w_i^* = w_i n_w / \sum_{i \in w} w_i$, so no wave dominates.

Table 1: Sample by wave, with the design-weighted percent favorable and its design-based standard error.

Wave	Respondents	Communes	Villages	% favorable	SE
1	1674	122	366	52.7	1.8
2	1672	122	366	54.9	2.2
3	1690	121	365	55.6	1.8

2.1 Diagnostics before modeling

Table 2: Computed before any model is fit. K: commune-wave cells. A: share of violence variation that region and wave dummies absorb; absorbed variation cannot inform the violence estimate. P: villages per commune-wave. W: within-commune share of violence variation.

Diagnostic	Meaning	Value	Status
K	Commune-wave cells (share with at least one event)	365 (42%)	
A, region	Region-wave violence explained by region + wave dummies (Model 1)	76%	OK
A, commune	Commune-wave violence explained by region + wave dummies (Models 2 to 4)	8%	OK
P	Median villages per commune-wave	3	
W	Within-commune share of violence variation (communes surveyed in 2+ waves)	18% (65 communes)	OK

Region and wave dummies absorb 76% of the variation in region-wave violence, so a model using region-wave violence (Model 1) must estimate the violence effect from what is left, spread over at most 24 region-wave values. Commune-wave violence is a different story: the same dummies absorb only 8% of it, because communes in the same region and wave experience very different levels of violence (Figure 1).

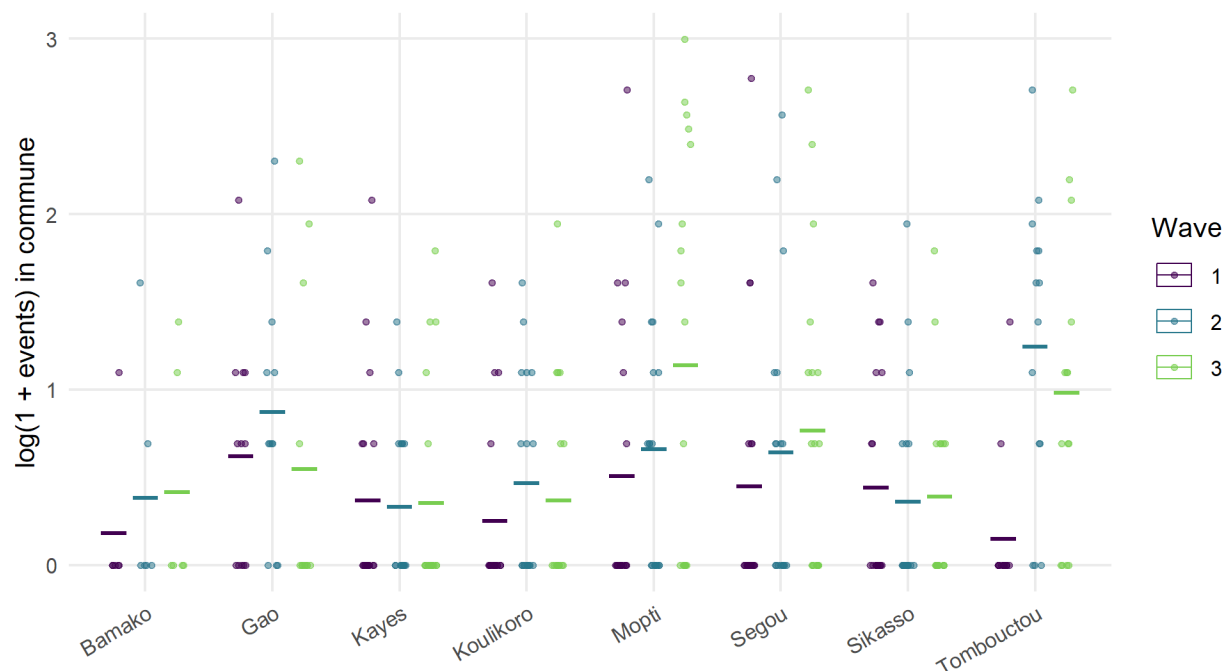


Figure 1: Commune violence within each region and wave. Each dot is a surveyed commune; the bar is the region-wave average. A region-wave measure gives every commune in a column the same value.

3 Models

3.1 Violence measure

For commune c in wave w , E_{cw} is the number of events and

$$V_{cw} = \log(1 + E_{cw}).$$

The log compresses a count that is zero for many communes and very large for a few; adding 1 keeps zero defined. On this scale, adding 1 multiplies $(1 + E_{cw})$ by $e \approx 2.72$, and adding $\log 2 \approx 0.69$ doubles it. Because these steps are hard to picture, results are also shown as predicted favorability at 0, 1, 5, and 20 events.

3.2 Equations

Respondent i lives in village v of commune c in region r and was interviewed in wave w . All models share the same base, with separate region and wave dummies and respondent covariates $\mathbf{x}_i$ (female, age per decade, ethnicity, education, urban):

$$\eta_i = \beta_0 + \sum_{r \neq \text{Bamako}} \alpha_r \text{Region}_r + \sum_{w=2}^3 \gamma_w \text{Wave}_w + \mathbf{x}_i' \boldsymbol{\beta}$$

Model	Linear predictor for logit $\Pr(y_i = 1)$	Clustering	Weights
M1	$\eta_i + \beta_R V_{rw}$	Survey design	Yes
M2	$\eta_i + \beta_V V_{cw}$	None	No
M3	$\eta_i + \beta_V V_{cw} + u_c$, with $u_c \sim N(0, \tau^2)$	Commune random in- tercept	No
M4	$\eta_i + \beta_V V_{cw}$	Survey design	Yes

Here $V_{rw} = \log\left(1 + \sum_{c \in S_{rw}} E_{cw}\right)$, where S_{rw} is the set of communes **surveyed** in region r and wave w (only surveyed communes have event counts). **M4 is the main model**. Each step changes one thing: M1 to M2 the violence measure, M2 to M3 the handling of clustering, M3 to M4 the move from a random effect to the survey design.

3.3 Estimation

M1 and M4 are survey-weighted logistic regressions (`survey::svyglm`). The coefficients solve the weighted score equations $\sum_i w_i^* \mathbf{x}_i (y_i - \mu_i) = 0$. Standard errors come from Taylor linearization using the first stage of the design (Binder, 1983):

$$\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}) = \hat{\mathbf{J}}^{-1} \left[\sum_h \frac{n_h}{n_h - 1} \sum_{j=1}^{n_h} (\mathbf{z}_{hj} - \bar{\mathbf{z}}_h)(\mathbf{z}_{hj} - \bar{\mathbf{z}}_h)' \right] \hat{\mathbf{J}}^{-1}, \quad \mathbf{z}_{hj} = \sum_{i \in (h,j)} w_i^* \mathbf{x}_i (y_i - \hat{\mu}_i)$$

where h indexes strata (region by wave), j communes, n_h the communes in stratum h , and $\hat{\mathbf{J}}$ the weighted information matrix. Commune totals $\mathbf{z}_{hj}$ include every village and respondent in the commune, so this also covers clustering within villages. Because violence is measured at the commune, the level where standard errors are computed is also the level where violence varies.

M2 is an ordinary logistic regression, included only to show what ignoring clustering does. **M3** is a multilevel logistic regression (`lme4::glmer`, Laplace approximation) without weights.

Reported effects. The average marginal effect (AME) is the design-weighted average change in the probability of a favorable view per one-unit increase in the violence measure:

$$\text{AME} = \frac{\sum_i w_i^* \partial \hat{\mu}_i / \partial V}{\sum_i w_i^*}$$

Predicted favorability at e events sets every respondent's commune to e events, keeps everything else as observed, and averages with the weights: $P(e) = \sum_i w_i^* \hat{\mu}_i(V = \log(1 + e)) / \sum_i w_i^*$. M3's AME is for a typical commune ($u_c = 0$), so it is close to, but not exactly comparable with, the population-averaged AMEs of the other models.

4 Results

Table 3: Main model (M4): predicted percent with a favorable view if every commune had the given number of events, other characteristics as observed (design-weighted, 95% CI).

Events in commune	Predicted % favorable	95% CI
0	58.3	55.9 to 60.7
1	52.7	50.6 to 54.8
5	43.8	40.0 to 47.6
20	34.1	27.8 to 40.3

Main result. If every commune had no events, 58.3% of respondents would be predicted to view the leader favorably; with 5 events, 43.8%. On the model scale, a one-unit increase in $\log(1 + E)$ changes the probability of a favorable view by **-7.9 percentage points** (95% CI -10.4 to -5.5), or about -5.5 points each time $(1 + E)$ doubles.

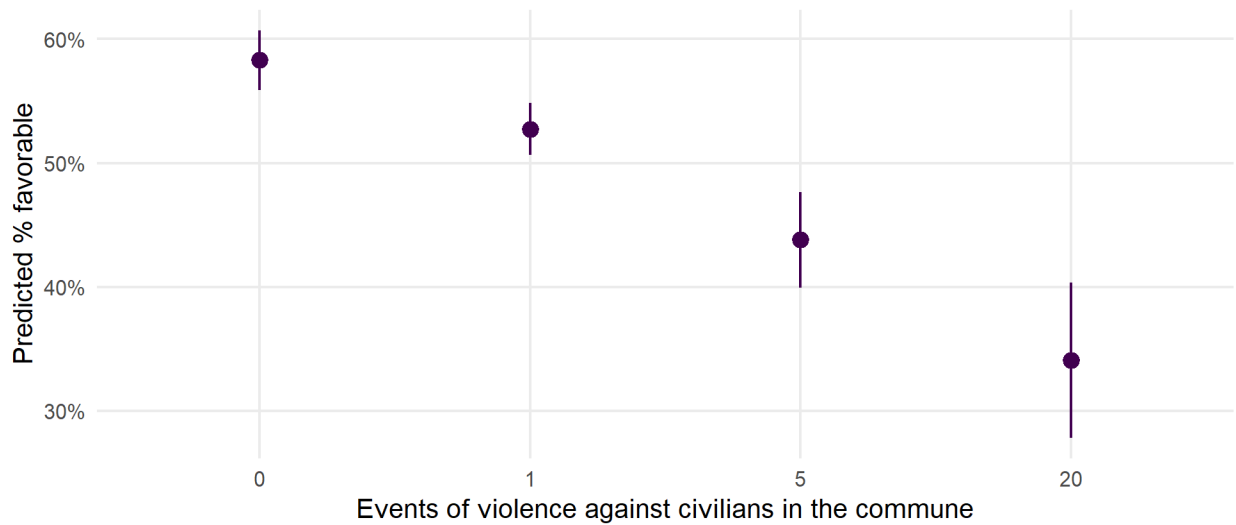


Figure 2: Main model (M4): predicted percent favorable at 0, 1, 5, and 20 events of violence against civilians in the commune (95% CI).

Table 4: Violence AME in percentage points per one-unit increase in the log violence measure, with 95% CI and standard error.

Model	AME [95% CI]	SE
M1 region-wave	-1.7 [-7.3, 4.0]	2.9
M2 naive	-7.6 [-9.4, -5.7]	0.9
M3 multilevel	-7.7 [-9.7, -5.8]	1.0
M4 survey (main)	-7.9 [-10.4, -5.5]	1.3
M4 + attitudes	-6.6 [-9.2, -4.0]	1.3

- **Model 1** gives -1.7 points (95% CI -7.3 to 4.0). With so little region-wave variation left after the dummies (Table 2), it cannot distinguish an effect from no effect.
- **Clustering.** M2 ignores clustering; the M4 standard error is 1.3 times as large. The wider M4 interval is the honest one.
- **Attitudes.** Adding local government approval, support for democracy, perceived violence, and stress (M4 + attitudes) shows how much of the association overlaps with these attitudes. Violence may itself change them, so this row is descriptive, not the main estimate.

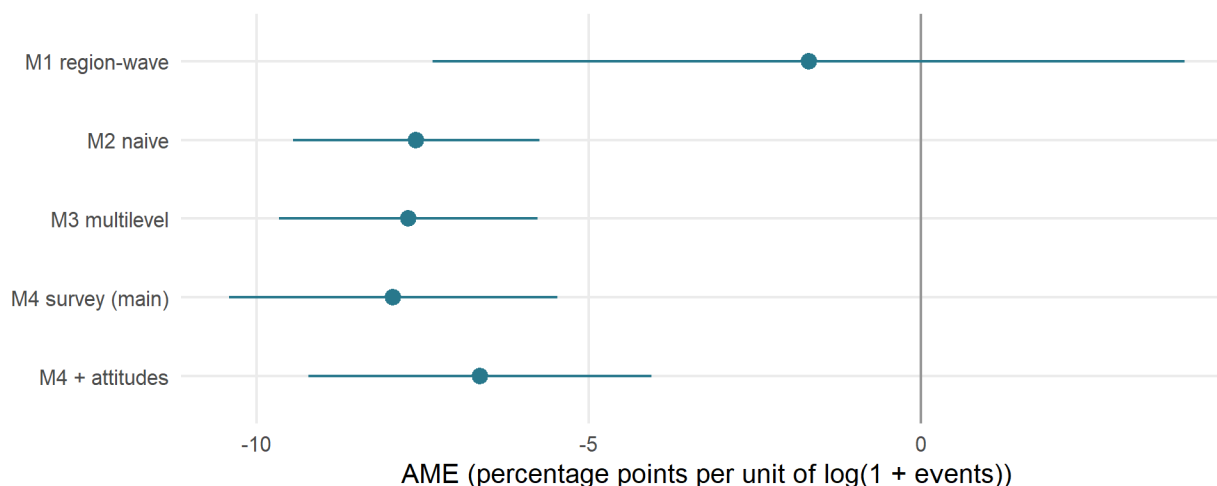


Figure 3: Violence AME across models (percentage points, 95% CI).

5 Maps

Commune values rest on few respondents each, so read the maps for broad patterns. Gray communes were not surveyed in that wave; this is missing data, not zero. Region names are shown on the Model 3 map.

Violence against civilians by commune and survey wave

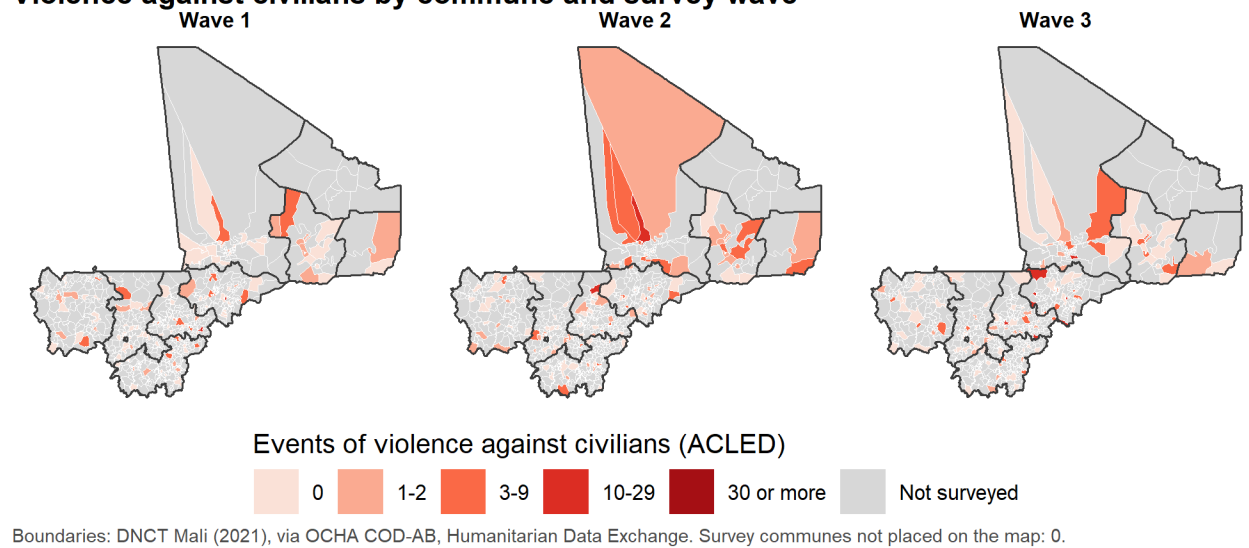


Figure 4: Events of violence against civilians by commune and wave (surveyed communes only).

Leader favorability by commune and survey wave

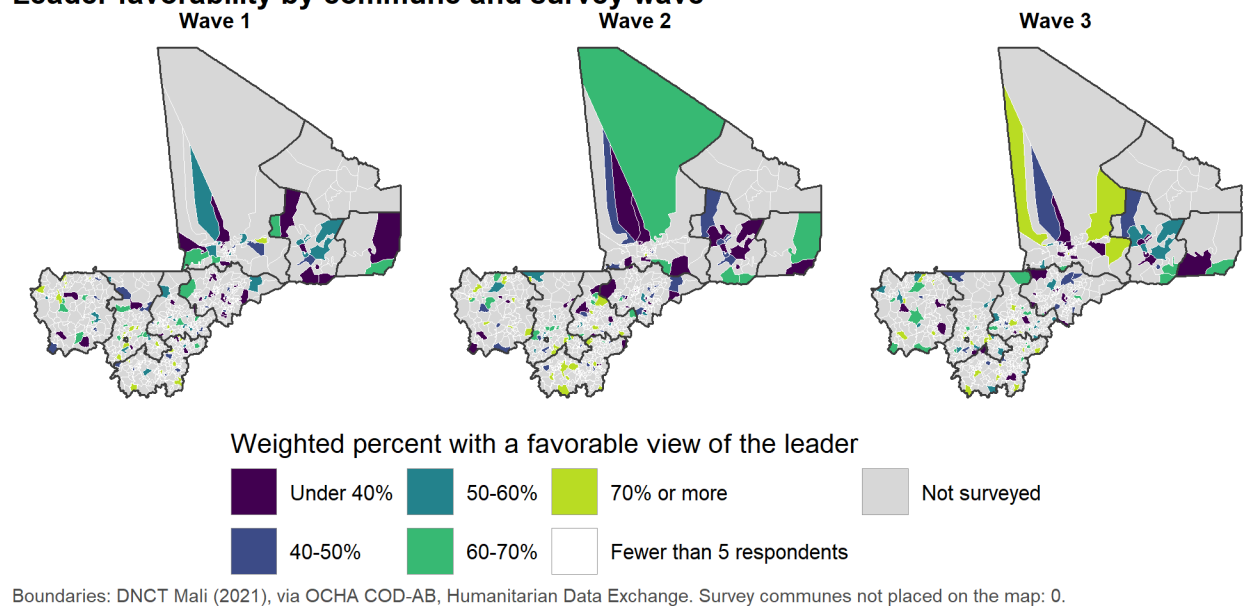
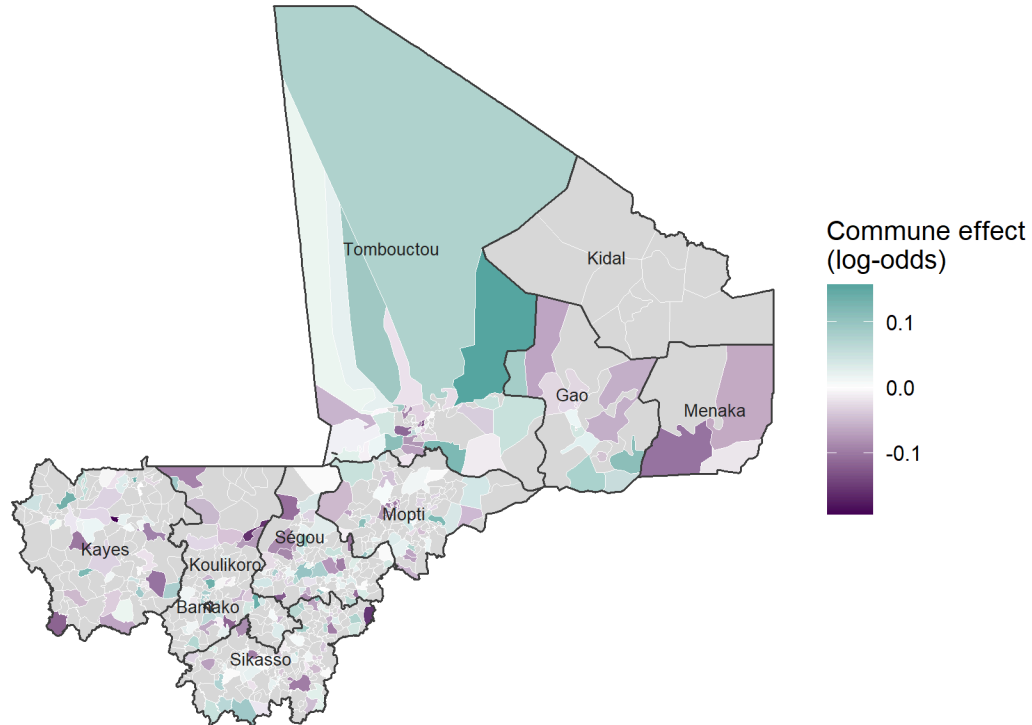


Figure 5: Design-weighted percent favorable by commune and wave.

Communes more or less favorable than Model 3 predicts



Gray: not surveyed. Boundaries: DNCT Mali (2021), via OCHA COD-AB, Humanitarian Data Exchange.

Figure 6: Model 3 commune effects: communes more (green) or less (purple) favorable than the model predicts from region, wave, violence, and respondent characteristics. Large same-colored areas would suggest a geographic factor the model misses.

6 Model checks

Model 3 converged without warnings; its commune variance is $\hat{\tau}^2 = 0.032$ (latent-scale ICC 0.010).

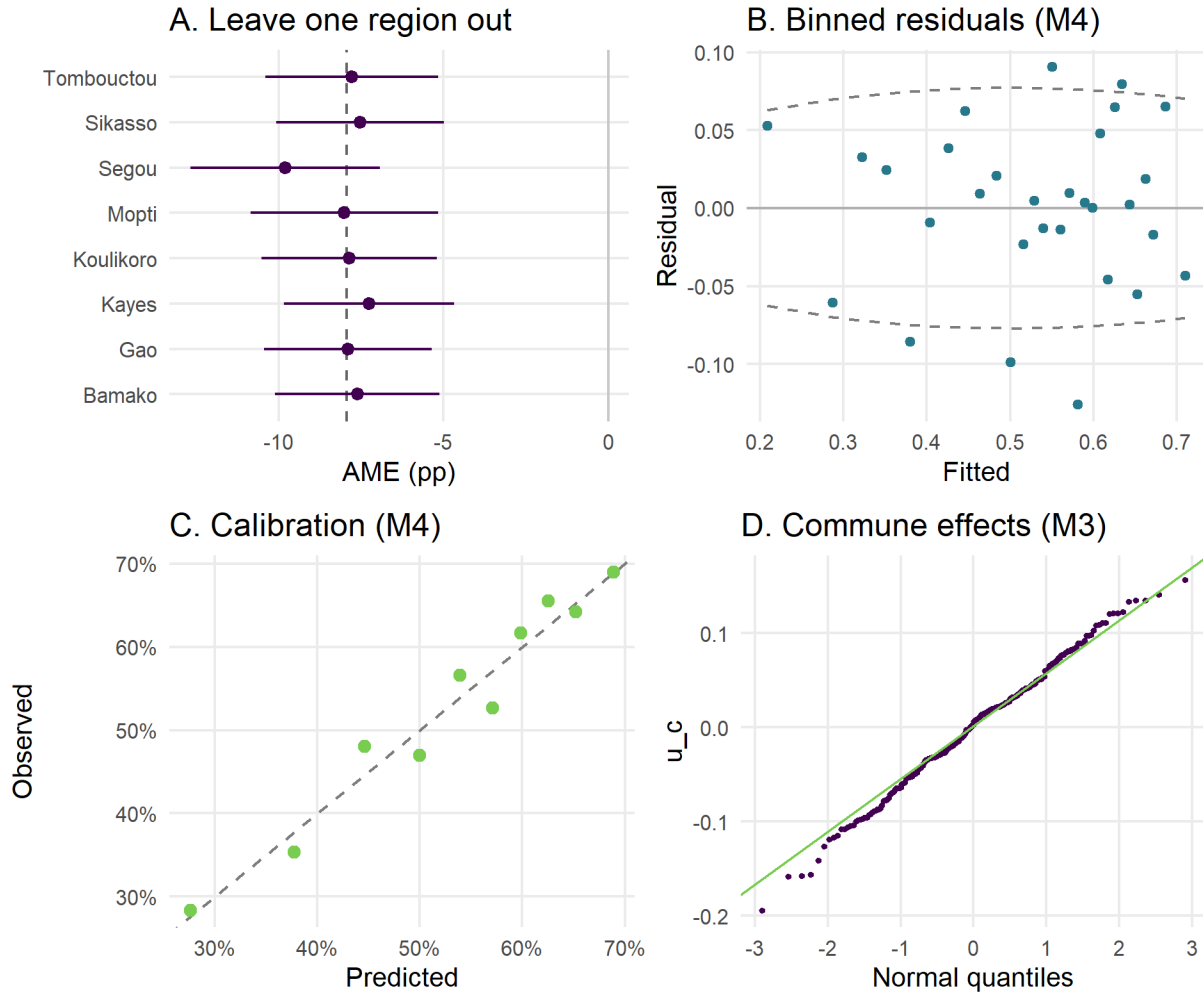


Figure 7: (A) M4 AME with each region dropped in turn; dashed line = full sample. (B) M4 weighted binned residuals; points should fall inside the dashed bounds with no pattern. (C) M4 calibration by decile. (D) M3 commune effects against a normal distribution.

7 How to explain the result

Among people surveyed in the same region and survey round, those living in communes with more violence against civilians were less likely to view the leader favorably. If every commune had experienced no such events, about 58.3% would be expected to hold a favorable view; with 5 events, about 43.8%. This compares different communes, so it shows an association, not proof that violence caused the change.

8 Limitations

- **Comparison between communes.** Most communes were surveyed once, so the estimate mainly compares different communes, not the same commune over time (W in Table 2). Stable differences between communes that the covariates miss (remoteness, local history) could drive part of it. The within-commune estimate is in the appendix.

- **Region-wide shocks.** Region and wave enter separately, so a shock that hit one region in one wave is not absorbed.
- **Violence measure.** Only surveyed communes have event counts. ACLED undercounts events in remote areas, which most likely weakens the association. The event window should end before fieldwork.
- **Coverage.** Insecure areas are harder to survey, so estimates describe the population that could be reached.
- **Design assumption.** The design is specified as communes (first stage) then villages. If communes were not a sampling stage, clustering at the commune is still appropriate because violence is measured there.
- **Map placement.** Communes are placed on the map through a reviewed name crosswalk. Names that could not be placed are counted in each map caption.

9 Appendix: within and between communes

Table 5: M4 with violence split into the commune's average across its surveyed waves (between) and that wave's deviation from it (within). AME in percentage points, 95% CI.

Part	AME [95% CI]	SE
Between communes (usual level)	-7.7 [-10.3, -5.0]	1.4
Within commune (change over time)	-9.6 [-16.0, -3.2]	3.3

The within part uses only communes surveyed in more than one wave whose violence changed; a commune surveyed once has a within value of 0. With little within-commune variation (18% here), the within estimate is imprecise and serves as a check, not a headline.

10 References

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